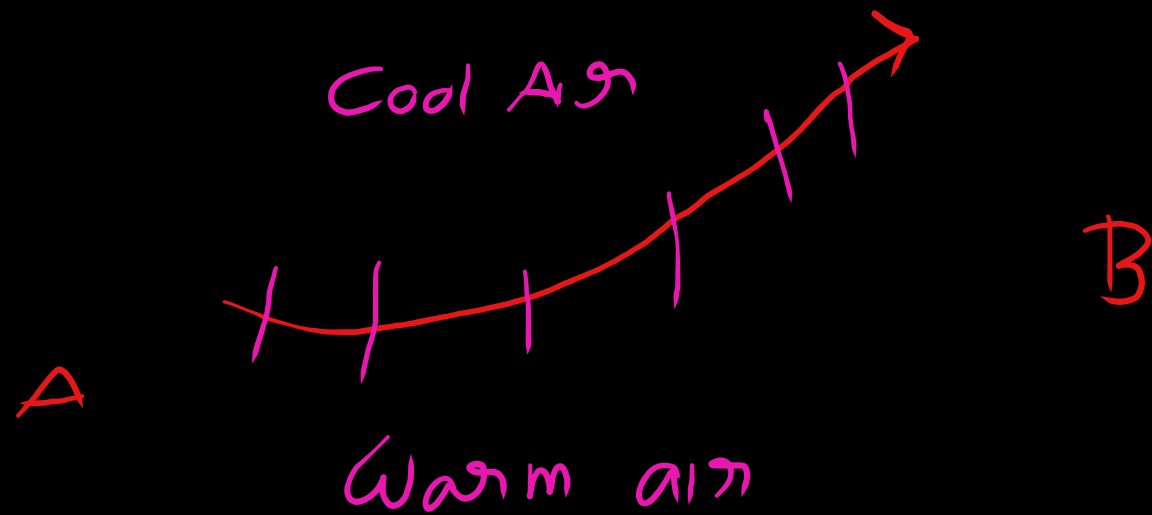


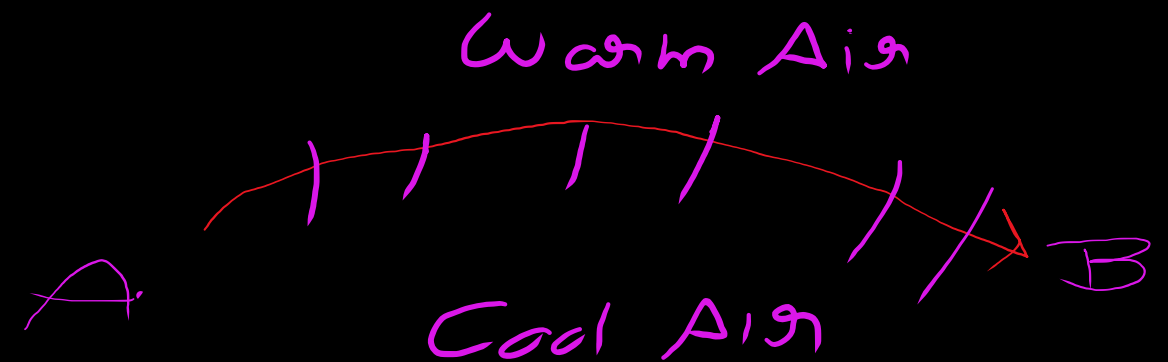
Refraction of Sound Waves

During night music/noise feels louder due to refraction of Sound Waves

Day



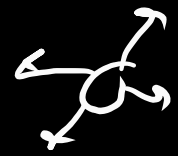
Night



Diffraction of Sound waves

→ Bending of waves through edges or holes is Diffraction

Diffraction

Light	Low	straight ✓	eg → Night highway Room X
Sound	High		Room ✓

Room 1 wall Room 2

Radio Sound ✓

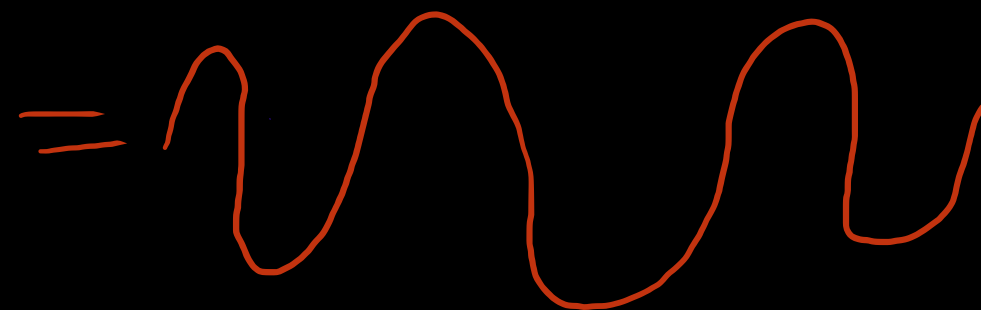
Torch Light X

Interference of Sound

Overlapping of Sound waves

Same ✓ ~~80%~~
diff ✗
No effect
on wave ✗

Speaker 1
+
Speaker 2



Same phase
↑ amplitude
or
↓ Opp. phase

→ Beats of music is due to interference

→ Near to Radio Station sometimes radio signals are not clear due to interference of waves

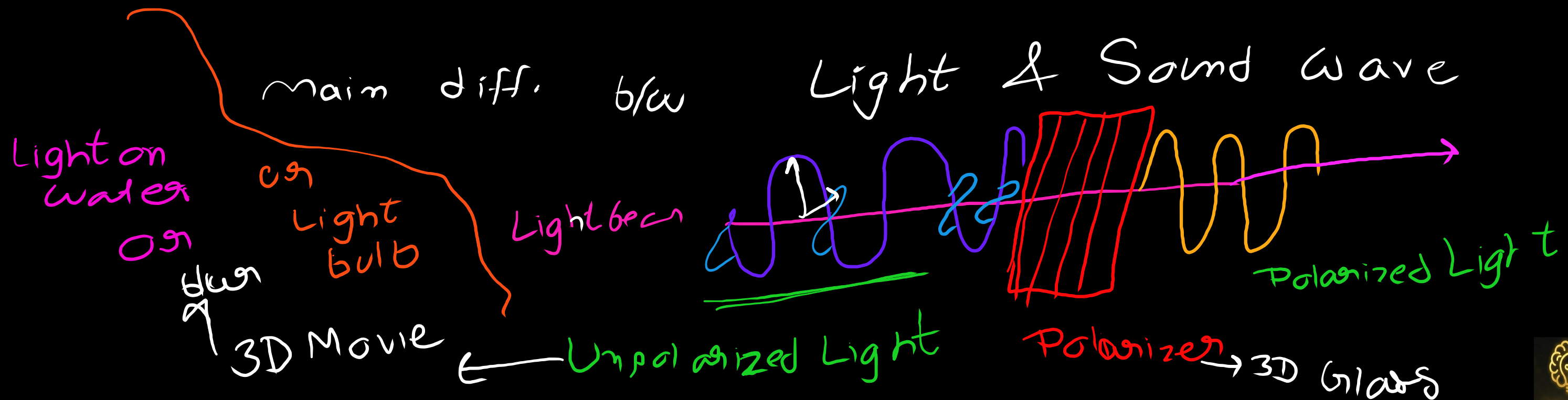
(Creation of Silence Zone)

multi plane
└
Single Plane

Polarization

Shown by Transverse waves Only $\theta \rightarrow 90^\circ$

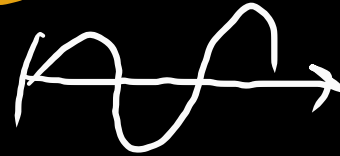
→ Some waves are longitudinal so they do not show Polarization.



Doppler Effect

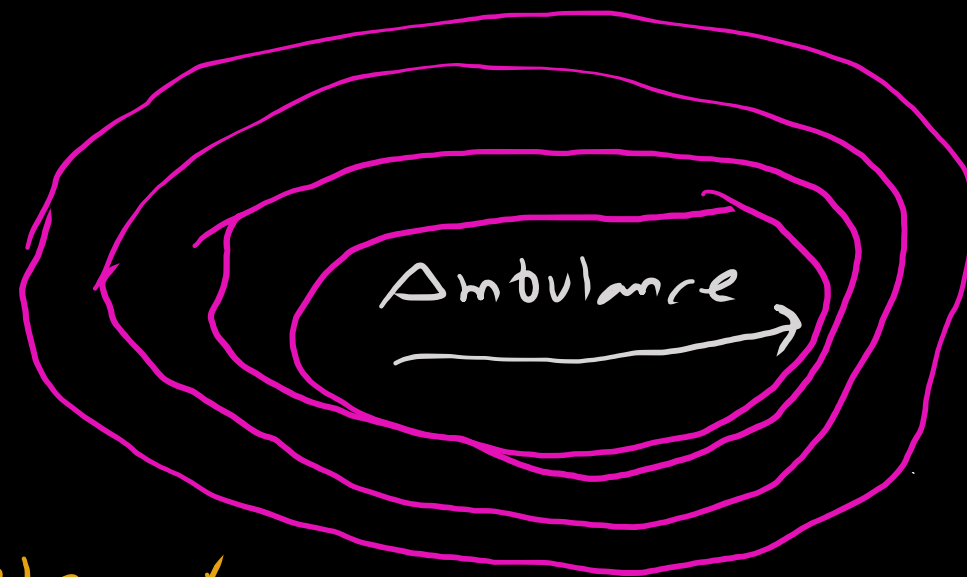
→ Shift in frequency of wave in relation to an observer due to relative motion of the wave src & observer

Low Freq



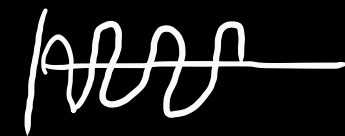
A

High wavelength



B

High Freq



Low wavelength